

EE225, Unveiling the Semiconductor World
Assessment - 2, Total Marks – 30, Weightage – 30%
Max. Time: 1.5 - 2 hrs.
Date – 26/04/2026

(The answer to each of these questions should not be more than half a page or a paragraph)

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- Q1: (2 marks)** The basic concept of the MOS transistor (Field Effect Transistor) was invented (patented) in 1926 by Lilienfeld, so why did it take almost 50 years before MOSFETs were introduced in manufacturing?
- Q2: (3 marks)** Draw and label the layout (not schematic diagram) of a MOS transistor with all four terminals (gate, source, drain, and body). Distinguish between the electrically active area of the device and the total area occupied on the wafer. How does the total device footprint change as the gate length is increased?
- Q3: (3 marks)** What is meant by the universal curve of a semiconductor technology? Explain its physical significance. Describe how the universal curve can be experimentally obtained.
- Q4: (3 marks)** At what technology nodes did the semiconductor industry transition from planar CMOS to FinFET, and subsequently toward Gate-All-Around (GAA) devices? Explain the primary scaling challenges that motivated the adoption of these advanced device architectures.
- Q5: (4 marks)** How are GPUs different than CPUs? What are streaming multiprocessors, and how do they contribute to parallelism? How did CUDA simplify GPU programming compared to early GPGPU approaches?
- Q6: (3 marks)** Distinguish between 1D, 2D, 2.5D, and 3D packaging technologies in semiconductor systems. Clearly explain the key features of each approach.
- Q7: (2 marks)** Discuss the role of interposers and hybrid bonding in advanced packaging.
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- Q8: (3 marks)** Explain the structure of a NAND flash cell array. Describe the Program and Erase operation. Why does this architecture enable high density and low cost compared to other memory types?
- Q9: (2 marks)** What is High Bandwidth Memory (HBM), and why is it critical for modern AI systems?
- Q10: (3 marks)** Write down the key stages for the CMOS technology development flow.
- Q11: (2 marks)** What do you mean by Bandwidth of a circuit? Why does many of the circuits could not work properly at Radio Frequencies?